

Stochastic Evolutionary Game Theory

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Synopsis

This work analyzes evolutionary strategies in finite populations through a stochastic birth-death type dynamics, using the Moran process and the Fermi function. We calculate the fixation probability and mean fixation time of a mutant, and connect this dynamics with the replicator equation. We study both the stochastic and deterministic regimes, identifying the four classical evolutionary scenarios. The analysis extends to fluctuating environments, where we explore the dilemma between specialized and generalist strategies, showing that the rate of environmental change can favor riskier behaviors. Finally, we apply this framework to tumors, showing how cellular interactions guide evolution and suggesting therapies that modify the environment rather than directly attacking cancer cells.

I. INTRODUCTION

In the early twentieth century, John von Neumann developed a branch of mathematics aimed at analyzing strategic decisions in contexts of interaction between multiple rational agents. Von Neumann defined classical game theory as “a theory that deals with the determination of optimal strategies in competitive games, where the outcome for each player depends on the choices of all participants” Von Neumann and Morgenstern (1944).

For its part, evolutionary game theory advocates for the study of strategies that lead to greater success in interaction within a population that tend to prevail, either through reproductive capacity or through imitation of successful behaviors, without assuming rationality or conscious decision-making: “evolutionary game theory is the application of game theory to populations evolving within biology. It defines a framework of competitions, strategies and analytical tools through which Darwinian competition can be modeled” Maynard Smith (1982).

The cornerstone of interactions between agents is the payoff matrix, which defines the benefit that each agent receives when facing different combinations of strategies:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad (1)$$

The value a represents the gain obtained by a player using strategy A when facing another player who also uses A ; b is the gain of A when playing against a player who employs strategy B ; c corresponds to the gain of B when facing a player with strategy A ; and finally, d is the gain received by a player with strategy B when playing against another player who also uses B , Traulsen and Hauert (2009).

To analyze how strategies evolve within a population, both replicator dynamics and stochastic dynamics are used. To relate both, it is fundamental to consider the limit of large populations to recover the deterministic dynamics.

II. STOCHASTIC EVOLUTIONARY DYNAMICS

We consider a system of N individuals in which all interact with each other. Suppose that j of them adopt strategy A and the remaining $N - j$ adopt strategy B . The probability of

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randomly selecting a pair formed by an individual with strategy A and another with strategy B is $\frac{j(N-j)}{N^2}$. To model the probability that an individual adopts the strategy of another, a stochastic rule based on the Fermi distribution is used, which depends on the difference in payoffs $\Delta\pi = \pi_A - \pi_B$ between strategies. The total payoff received by an individual with strategy A and another with strategy B is respectively: $\pi_A(j) = \frac{j-1}{N}a + \frac{N-j}{N}b$ and $\pi_B(j) = \frac{j}{N}c + \frac{N-j-1}{N}d$ Calvo, Muñoz, and Galla (2025).

- The probability that an individual with strategy B adopts strategy A (i.e. $j \rightarrow j+1$):

$$T_j^+ = \frac{j(N-j)}{N^2} \cdot \frac{1}{1+\exp(-w\Delta\pi)}$$
- The probability that an individual with strategy A adopts strategy B (i.e. $j \rightarrow j-1$):

$$T_j^- = \frac{j(N-j)}{N^2} \cdot \frac{1}{1+\exp(w\Delta\pi)}$$

The concepts of fixation probability and fixation time of a mutant are derived to understand its possible evolutionary impact on a finite population with absorbing states $j=0; j=N$, Traulsen and Hauert (2009).

$$\text{Fixation probability of a mutant} \quad \phi_1 = \frac{1}{\left(1 + \sum_{m=1}^{N-1} \prod_{j=1}^m \xi_j\right)} \quad (2)$$

$$\text{Fixation time of a mutant} \quad t_1 = \phi_1 \left(\sum_{w=1}^{N-1} \sum_{l=1}^w \frac{1}{T_l^+} \prod_{m=l+1}^w \xi_m \right) \quad (3)$$

All mathematical development is found in Appendix A.

A. Interpretation of birth-death dynamics

Starting from the stochastic dynamics defined by the transition probabilities $T_j^\pm$, it is possible to interpret the process as a birth-death dynamics in discrete time for the frequency $x = j/N$ of strategy A . This interpretation allows the application of the master equation formalism and, through the Kramers-Moyal expansion, approximate the dynamics by a Langevin equation considering the drift term $A(x)$ and diffusion term $B(x)$, Ibáñez (b).

$$\dot{x}(t) = A(x) + \sqrt{B(x)}\xi(t) \Rightarrow \dot{x}(t) = (T_x^+ - T_x^-) + \sqrt{\frac{(T_x^+ + T_x^-)}{N}}\xi(t) \quad (4)$$

Taking the limit $N \rightarrow \infty$ we recover the replicator dynamics:

$$\dot{x}(t) = x(1-x)\tanh\left(+\frac{w}{2}\Delta\pi(x)\right) = x(1-x)\tanh\left(+\frac{w}{2}[\pi_A(x) - \pi_B(x)]\right) \quad (5)$$

All mathematical development is found in Appendix B.

B. Stability of fixed points

We consider the system $\dot{x}(t) = x(1-x)\tanh(+\frac{w}{2}(\alpha x + \delta))$, where $\alpha = (a - b - c + d)$, $\delta = (b - d)$. The fixed points x^* are found by solving the equation $\dot{x} = 0$: $x^* = 0$, $x^* = 1$, or $x^* = \frac{-\delta}{\alpha}$.

Stability can be analyzed through the derivative $f'(x^*)$. The adjusted replicator equation consistently reproduces the same stability conditions and nature of fixed points as the traditional replicator equation. The four generic cases of evolutionary dynamics are analyzed: (1) Dominance, (2) Bistability, (3) Coexistence, and (4) Neutrality, Traulsen and Hauert (2009).

C. Fixation in fluctuating environments

We consider a mutant of type A introduced into a finite population of $N - 1$ residents of type B. The fact that the population is finite introduces “demographic noise”, an intrinsic stochasticity that allows phenomena such as fixation or extinction of the mutant Baron and Galla (2016). Another source of stochasticity is an environment (σ) that changes with a certain probability between two states. Examples of changing environments are found in external variations such as temperature, pH, presence or absence of nutrients, or in direct intervention such as antibiotic injection Kussell *et al.* (2005).

This environmental dependence can be easily introduced through the payoff matrix:

$$\begin{pmatrix} a_\sigma & b_\sigma \\ c_\sigma & d_\sigma \end{pmatrix} = \begin{pmatrix} 1 & 1 + 0.5 \cdot \sigma \\ 1 + 0.9 \cdot \sigma & 1 \end{pmatrix}$$

Where the population dynamics alternates between two possible states ($\sigma = \pm 1$). In the first case ($\sigma = 1, b_\sigma, c_\sigma > 1$), the environment tends to maintain both species in the population, since individuals of different categories obtain a higher payoff interacting with each other. Therefore, this dynamic is usually denoted as coexistence game. Conversely, in the second case ($\sigma = -1, b_\sigma, c_\sigma < 1$), evolutionary dynamics favor homogeneity, and this dynamic is usually denoted as coordination game (or bistability).

We study the reproductive success of a mutant (type A) introduced in a population of $N - 1$ residents (type B) in an environment that changes with probability p_+ from coexistence to coordination, and with probability p_- of changing from coordination to coexistence.

Evolutionary dynamics follow a Moran process, where at each step, based on fitness, a model individual reproduces and a target individual dies. We take the fitness function as an exponential: $f_A^s(j) = e^{\beta \pi_A^s(j)}$ and $f_B^s(j) = e^{\beta \pi_B^s(j)}$. Through the parameter $\beta > 0$ we can calibrate the intensity of selection.

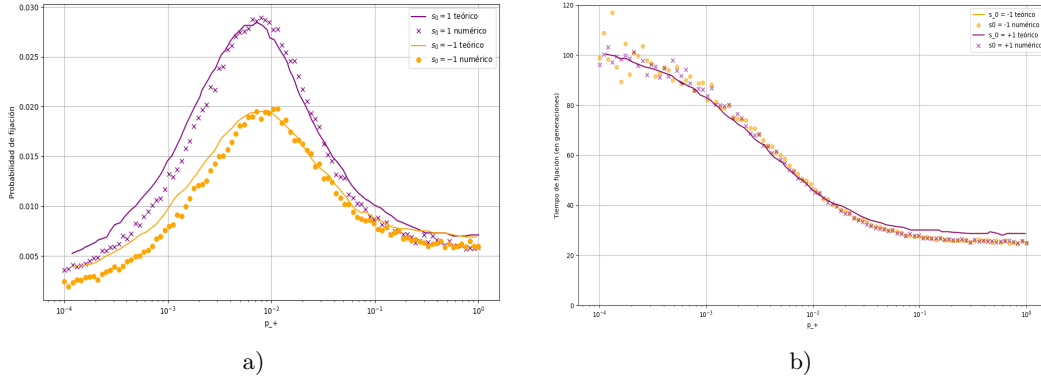


FIG. 1: a) Fixation probability of a mutant as a function of p_+ for two different initial environments ($\sigma \pm 1$) with $p_- = 0.01$. b) Average fixation time (in generations) of a mutant as a function of p_+ for two different initial environments with $p_- = 0.01$.

For small values of p_+ , the environment tends to remain in its initial state for a long time. No notable difference is observed between starting at $\sigma = +1$ or at $\sigma = -1$. However, high values of p_+ cause the environment to fluctuate more rapidly. Both alternation that is too slow and alternation that is too fast harm absorption, therefore a maximum of fixation probability is observed around $p_+ \approx p_-$.

D. Neutral selection under the fast switching limit

We proceed to analyze a particular case of payoff matrix under the context of the fast switching limit. This limit models situations where the interactions between agents change

rapidly compared to the time scale of strategy evolution.

We consider the following payoff matrix:

$$M = \begin{bmatrix} b & b \\ 1 & 1 \end{bmatrix} \quad (6)$$

We substitute each coefficient in the replicator equation obtaining $\dot{x} = x(1-x)[(b-1)]$. For fixed points we distinguish between cases where $b = 1$ (neutral selection) and $b \neq 1$ (where fixed points are $x^* = 0$ and $x^* = 1$).

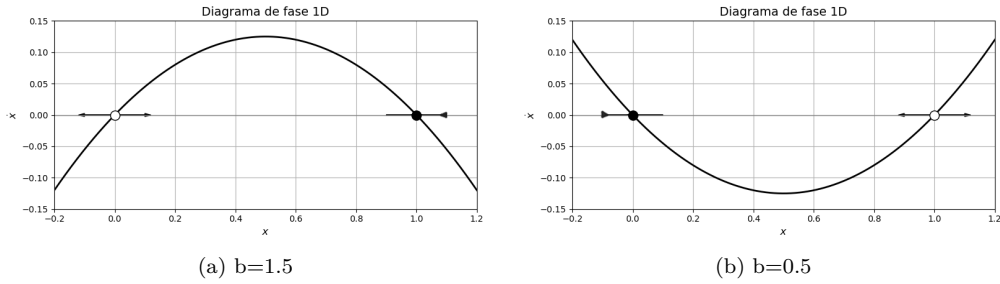


FIG. 2: One-dimensional phase diagrams for replicator dynamics $\dot{x} = x(1-x)(b-1)$ with different values of the parameter b . The fixed points at $x = 0$ and $x = 1$ are indicated with black (stable) or white (unstable) circles.

We extend the analysis to the case in which the values b fluctuate randomly at each step according to the probability distribution $f(b)$. The effective transition probability is defined as:

$$T_{j,\text{eff}}^{\pm} = \frac{j}{N} \cdot \frac{N-j}{N} I^{\pm} \quad (7)$$

The expression of the neutral selection line under the fast switching limit can be written as:

$$\mu = 1 - \sigma \sqrt{\frac{p_1}{p_2}} - \frac{1}{\beta} \ln f(\sigma) \quad (8)$$

E. Risk-averse or risk-prone strategies?

Taking as primitive properties in the study of fixation probability the average payoff of the mutant (μ) and the variance of payoff (σ^2), we ask under what conditions it is convenient for the mutant to “play it safe” —adopt a strategy with lower variance— or to “bet” —adopt a strategy with higher variance. Environmental fluctuations can induce an advantage for strategies with higher variance, even when these have a lower average payoff.

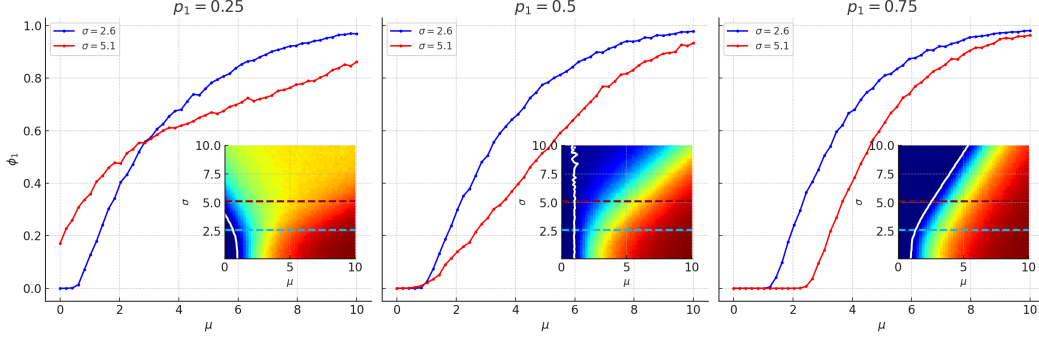


FIG. 3: Fixation probability as a function of the average payoff value of the mutant μ for $\sigma = 5.1$ (red) and $\sigma = 2.6$ (blue). The scale ranges from blue (probability close to 0) to red (probability close to 1).

When the mutant's payoff is lower on average than the resident ($\mu < 0$), strategies with higher variance (risk) obtain higher fixation probability: the bet is safe. The mutant can specialize in the favorable environment even if it means lower performance in the unfavorable environment.

The case of fast switching only allows us to distinguish two regimes: betting or playing it safe. However, if instead of considering the fast switching limit we analyze the general case where the environment changes with certain probability, we can specify the type of bet.

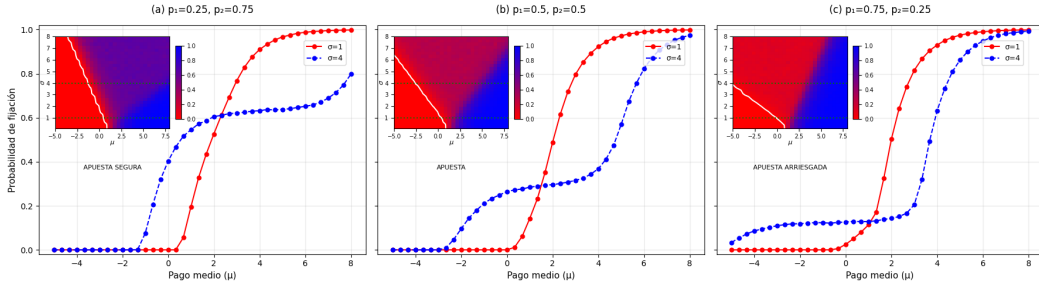


FIG. 4: Fixation probability of a mutant for payoff distributions $p_1 = 0.25, 0.5$ and 0.75 . The main figures show fixation probability as a function of average payoff for $\sigma = 1$ and $\sigma = 4$.

Strategies of “playing it safe” (variance-averse) are advantageous for the mutant when its average payoff is higher than the resident's ($\mu > 0$): reducing part of its average payoff in exchange for less suffering from environmental change turns it into a generalist.

III. TUMOR DYNAMICS

Tumors represent a direct application example of game theory, since cancer cells have lost their normal cooperative behavior and defect. However, results indicate that tumors do not consist entirely of cancer cells, which raises the possibility that cysts constitute an equilibrium due to interactions between tumor and normal cells Gerstung, Nakhoul, and Beerenwinkel (2011).

In their study of multiple myeloma (MM), a type of bone marrow cancer, Pacheco, Santos, and Dingli (2014) analyze, from an evolutionary game theory approach, the equilibrium of bone remodeling, determined by the interaction between bone resorption —carried out by

osteoclasts (OC)— and bone formation —performed by osteoblasts (OB). In our framework, this is modeled as a coexistence game.

However, the appearance of MM cells alters this dynamic equilibrium between OB and OC in favor of OC, since MM cells secrete chemical precursor components of OC. Additionally, OC cells also stimulate the growth of MM and these latter secrete proteins that prevent the creation of new OB cells. This complex interaction can be described through a three-player payoff matrix.

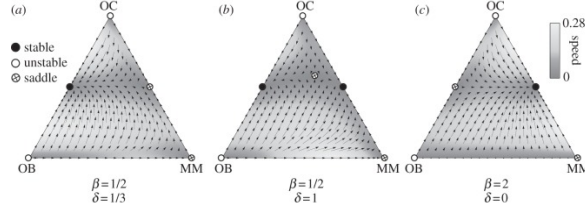


FIG. 5: Evolutionary dynamics of OC, OB, and MM. a) If $\beta < 1$: MM cells can go extinct, restoring OB-OC equilibrium. b) If $\beta < 1$ but $\beta + \delta > 1$, a saddle point appears. c) If $\beta > 1$, the only stable equilibrium is the coexistence of MM and OC, leading to bone destruction.

From a medical perspective, therapies that reduce β decrease lesions and slow cancer progression, while drugs that reduce δ improve bone mass and reduce the number of cancer cells.

IV. CONCLUSIONS

This work has explored how evolutionary game theory can serve as a rigorous framework for connecting mathematical models with observable biological phenomena. The equations developed formalize evolutionary principles relevant in different fields of biology, from molecular interactions to population and ecological dynamics.

The analysis of fluctuating environments has identified a non-trivial evolutionary dynamic: optimal adaptability is not achieved under conditions of complete stability or extreme variability, but in intermediate regimes. This observation raises a revision of biological fitness, highlighting its dependence on the fluctuating environment.

Adopting high-variance strategies (betting) or low-variance strategies (playing it safe) depends on environmental conditions and the average payoff relative to the resident. When the mutant presents lower average payoff ($\mu < 0$), betting can confer an advantage in fixation probability. Conversely, when the average payoff is higher ($\mu > 0$), reducing variance becomes advantageous.

The application of evolutionary game theory to tumor dynamics demonstrates how changes in the structure of cellular interactions can induce transitions between coexistence and coordination evolutionary regimes. This approach allows us to reinterpret cancer progression as an alteration of population balance, driven by parameters such as β and δ , and suggests that effective therapies could be oriented toward modifying these interactions rather than directly attacking tumor cells.

Finally, the developed framework offers relevant implications in different research areas. In oncology, it suggests therapeutic approaches based on modification of the tumor microenvironment. In ecology, it provides tools to model the viability of invasive species under non-stationary environmental conditions. And in evolutionary biology, it establishes a formal link between game theory and nonlinear dynamic models. Overall, this study contributes to a more integrated vision of evolution, where mathematical models allow descriptions of complex biological systems.

Appendix A: Derivation of expressions for stochastic fixation

1. Derivation of the fixation probability of a mutant

The fixation probability of a mutant is calculated. For an intermediate state, a balance equation is used taking into account all possible transitions in the next step: $\phi_j = T_j^+ \phi_{j+1} + T_j^- \phi_{j-1} + (1 - T_j^+ - T_j^-) \phi_j$.

The balance equation can be rewritten as: $0 = -T_j^-(\phi_j - \phi_{j-1}) + T_j^+(\phi_{j+1} - \phi_j)$.

Define $y_j = \phi_j - \phi_{j-1}$. Then: $0 = T_j^+ y_{j+1} - T_j^- y_j \Rightarrow y_{j+1} = \frac{T_j^-}{T_j^+} y_j = \xi_j y_j$.

Using the recursive relation and boundary conditions $\phi_0 = 0, \phi_N = 1$, we arrive at:

$$\phi_1 = \frac{1}{\left(1 + \sum_{m=1}^{N-1} \prod_{j=1}^m \xi_j\right)}$$

2. Derivation of the fixation time of a mutant

Similarly, define $z_j = t_j - t_{j-1}$. The recursive relation becomes:

$$z_{j+1} = \xi_j z_j - \frac{1}{T_j^+}$$

Using absorbing conditions $t_0 = 0, t_N = 0$ and summing, we obtain:

$$t_1 = \phi_1 \left(\sum_{w=1}^{N-1} \sum_{l=1}^w \frac{1}{T_l^+} \prod_{m=l+1}^w \xi_m \right)$$

Appendix B: Derivation of expressions for birth-death dynamics

1. Derivation of the Langevin equation

A birth-death dynamics is considered. We calculate the probability of having j individuals at time $t + \Delta t$. Using the Kramers-Moyal expansion with $\Delta x = 1/N$:

$$P(x; t + \Delta t) - P(x; t) = -\frac{\partial_x}{N} ((T_x^+ - T_x^-) P(x; t)) + \frac{\partial_x^2}{2N^2} ((T_x^+ + T_x^-) P(x; t))$$

Using the definition of partial time derivative and the relation $\Delta t \sim h$:

$$\partial_t P(x; t) = -\partial_x (A(x) P(x; t)) + \frac{1}{2} \partial_x^2 (B(x) P(x; t))$$

where $A(x) = (T_x^+ - T_x^-)$ and $B(x) = (T_x^+ + T_x^-)/N$.

The associated Langevin equation is:

$$\dot{x}(t) = A(x) + \sqrt{B(x)} \xi(t)$$

2. Derivation of the adjusted replicator equation

Taking the limit $N \rightarrow \infty$ to recover the adjusted replicator dynamics:

$$\dot{x}(t) = x(1-x)\tanh\left(+\frac{w}{2}\Delta\pi(x)\right)$$

Appendix C: Fixation in fluctuating environments

Fixation probability obeys boundary conditions $\phi_{0,\sigma} = 0$ and $\phi_{N,\sigma} = 1$. The recursive calculation involves the payoff matrix determinant and recursive terms extracted using appropriate constraints.

Appendix D: Fast switching limit derivations

1. Neutral selection condition

The neutral selection line is defined as the set of parameters (μ, σ) such that the fixation probability equals $1/N$. This occurs when $T_{j,eff}^- = T_{j,eff}^+$.

2. Neutral selection expression

In the fast switching limit, the mutant “feels” a weighted average of both environments. Starting from the condition $I^- = 1/2$, we obtain:

$$\mu = 1 - \sigma \sqrt{\frac{p_1}{p_2}} - \frac{1}{\beta} \ln f(\sigma)$$

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